

W09 B ONSITE NON-POTABLE WATER REUSE | O (MAX : 2 PT)

Intent : Conserve water through non-potable water systems without compromising the health of the building occupants.

Summary : This WELL feature requires projects to implement a safety plan when capturing and using non-potable water within the project boundary.

Issue : Efficient water management is an ever-growing need for supporting and sustaining human existence.¹ The rise of extreme droughts and subsequent long-term water scarcity are requiring changes in how humans think about and use water.¹ Conversely, severe rain events can overload wastewater treatment plants with storm- and rainwater, potentially resulting in combined sewer overflows (CSOs), in which untreated water is released to natural bodies of water and beaches downstream.² Buildings can reduce the pressure on city-wide water infrastructure by supporting efficient allocation of resources, such as minimizing the use of municipally sourced water for applications that do not require potability (e.g., irrigation, flushing toilets).² However, without proper design and operations management, there is a risk of contamination of potable water lines or oral or respiratory exposure to non-potable water.²⁻⁶

Solutions : The design and operations of a healthy, non-potable water system must address the water sources and uses. For instance, water coming from clothes washers and showers, as well as stormwater collected at the street level, requires treatment and water quality monitoring before reuse. The system should be designed so that water capture and reuse does not create niches of water stagnation, and storage tanks for non-potable water need periodic checks and safeguards against overflows.⁵ Additionally, pipes destined to distribute non-potable water must not be commingled with those for drinking water, and proper control measures must be in place if potable water supplements non-potable water applications (e.g., toilet flushing).² Finally, smells and odors need to be managed to avoid the perception of a health threat, as this may hinder efforts to improve water management and conservation in a building.^{7,8} While there are many factors to consider, these challenges can be managed through a properly implemented safety plan that addresses health endpoints in the collection, treatment and distribution of non-potable water within the building.⁹ The success in the implementation of this plan also depends on third party verification, in which the plan is evaluated with respect to the achievement of its health endpoints.¹⁰ Additionally, education and signage are essential in supporting occupants' understanding of the positive impact of sensible water utilization within buildings and of the measures taken to protect human health.¹¹ By properly managing and controlling non-potable water reuse and capture systems, projects can help maintain occupant health and safety, while contributing to decreased water use by up to 75% in new buildings.¹²

PART 1 IMPLEMENT SAFETY PLAN FOR NON-POTABLE WATER CAPTURE AND REUSE (MAX : 2 PT)

For All Spaces:

1: Safety plan description

The project or organization implements a safety plan that contains the following:⁵⁻⁹

- a. A list of key roles for design, operations, maintenance and third-party inspection of the non-potable water system capture, treatment and use.
- b. A list of all applicable codes and regulations in the jurisdiction where the non-potable water reuse system is being installed and that govern the design, commissioning, and approval of operation of the system.
- c. A process flow diagram that displays the non-potable water sources, conveyances, storage units, treatment devices and points of use, emphasizing the points where makeup potable water (i.e., water needed to supplement non-potable needs) may be added.
- d. A description of the system that includes the sources and estimated contaminant loads of the non-potable water, the intended uses for the non-potable water, the water treatment devices (if any) and their certifications, and the water quality parameters expected at the points of use.
- e. An analysis of how human exposure to pathogens through ingestion and inhalation of non-potable water is minimized, including (if applicable) a description of how the potable water network is protected from the introduction of non-potable water, emphasizing strategies that address cross-connection control and backflow prevention.
- f. A description of the signage and identifiable pipe color-coding to distinguish the non-potable water network.
- g. A list of strategies for the control of odors, nuisances and vectors due to stagnation of non-potable water.
- h. The provisions for emergency operations caused by overflow of storage tanks, leaks and outages.
- i. A list of operational parameters (such as flow, turbidity, coliforms or other treatment-dependent indicators) to monitor the intended functioning of the water system, their monitoring frequency and control actions if such parameters are beyond target ranges.
- j. A list of control points where the operational parameters are being measured.
- k. A list of routine maintenance protocols and schedules.
- l. A description of the procedures for system startup, determination of protocols for verification of the safety plan, including Legionella testing if a risk of inhalation exists, and schedule for third-party inspections.

2: Safety plan implementation

Documentation of the following are submitted annually through the WELL digital platform:

- a. Startup procedure.
- b. Maintenance logs.
- c. Results from verification tests (when applicable).
- d. Third-party inspections.

3: Safety plan communications

Conspicuous signage is present to communicate the following:

- a. To distinguish potable from non-potable water (where applicable).
- b. To highlight the safety features and conservation goals of the non-potable water system.

Note :

Projects are not required to provide information on the individuals assigned to perform the key roles determined in the safety plan.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

REFERENCES

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